

REGIONAL SALES PERFORMANCE

ANALYZING AND VISUALIZING

Project Title : Analyzing and Visualizing Regional Sales Performance

Name : Hariharan S

Domain : Retail Sales Analytics

Tools: Microsoft Excel

Problem Statement:

ABC Electronics wants to assess its sales performance across different regions and product categories. The management needs a summarized report with visual insights to identify top-performing regions, products, and trends. As a business analyst, your task is to clean the data, analyze it using formulas, and create a basic dashboard that provides actionable insights.

Business Use Cases:

- Identifying **top-performing regions** and product categories.
- Understanding the **impact of discounts** on sales and profit margins.
- Providing **actionable insights** for marketing and inventory management.
- Monitoring key sales metrics for **strategic decision-making**.

Data Set: Sales Performance Analysis

Data	Description
Order ID	Unique Identification Number
Order Date	Order Placed Date
Region	Region Wise Sales(North,South&etc)
Product Category	Type based product (electronics & etc)
Sales Amount	Revenue generated
Quantity Sold	Units sold
Discount	Discount applied
Profit	Profit earned

Data Set Explanation:

The dataset contains 4000 rows of sales records for ABC Electronics. Each record includes details such as order ID, date, region, product category, sales amount, quantity sold, discount percentage, and profit. You are required to preprocess the dataset (e.g., cleaning, merging) and perform data analysis.

Tools Used:

Microsoft Excel:

Used for data cleaning, analysis, calculations, and reporting.

Features:

- **Pivot Tables** – Used to summarize and analyze large datasets dynamically.
- **Excel Charts** – Used to visualize sales trends and performance insights.
- **Conditional Formatting** – Used to highlight important patterns and high-performing data.
- **Regression Analysis** – Used to study the relationship between discounts and sales performance.
- **Dashboard** – Used to create an interactive and user-friendly sales performance dashboard.

Approach:

The project follows a systematic approach to analyze and visualize regional sales performance using Microsoft Excel.

1. Data Collection

The sales dataset containing order details, region, product category, sales amount, discount percentage, and profit was collected for analysis.

2. Data Cleaning and Preprocessing

- ✓ Removing unwanted spaces using TRIM
- ✓ Standardizing text using UPPER and LOWER functions
- ✓ Checking missing or duplicate values
- ✓ Formatting dates and numeric fields properly

	A	B	C	D	E	F	G	H	I	J	K
1	Order ID	Order Date	Region	Product Category	Sales Amount	Quantity Sold	Discount	Profit	Order ID - TRIM	Region - UPPER	Product Category - Lower
2	ab899d2f	2023-04-06	North	Groceries	859.29	44	37	1653.87	ab899d2f	NORTH	groceries
3	73484eb1	2024-04-30	West	Groceries	1546.81	23	33	1932.71	73484eb1	WEST	groceries
4	19395924	2024-03-12	East	Groceries	4703.58	31	15	527.47	19395924	EAST	groceries
5	b3fde3d	2024-02-09	East	Groceries	3377.17	47	3	482.65	b3fde3d	EAST	groceries
6	6e4bdd51	2024-07-29	South	Clothing	3517.49	6	28	748.72	6e4bdd51	SOUTH	clothing
7	21d88d69	2023-08-26	East	Furniture	362.55	19	50	314.71	21d88d69	EAST	furniture
8	9da2704d	2024-10-07	East	Sports	4811.17	22	18	1372.89	9da2704d	EAST	sports
9	a1bd6815	2023-03-15	West	Electronics	3565.2	22	16	1437.3	a1bd6815	WEST	electronics
10	0bbd51fc	2023-06-29	East	Clothing	3805.25	19	3	1537.28	0bbd51fc	EAST	clothing

3. Using Excel Formulas

- ✓ SUM for total sales
- ✓ AVERAGE for profit and discount analysis
- ✓ Logical and statistical functions for business insights
- ✓ Profit Calculation by the product Quantity Sold.

Profit	Average_by_region	Average Discount by Products(%)	Average Profit by Region
1653.87	2573.119554	24.47	983.09
1932.71	2535.998906	24.47	1,021.30
527.47	2507.726068	24.47	1,016.10
482.65	2507.726068	24.47	1,016.10
748.72	2494.41128	24.89	1,010.90
314.71	2507.726068	25.23	1,016.10
1372.89	2507.726068	24.81	1,016.10
1437.3	2535.998906	24.45	1,021.30
1537.28	2507.726068	24.89	1,016.10

4. Pivot Table Analysis

- ✓ Total sales by region
- ✓ Profit by product category
- ✓ Region-wise performance comparison
- ✓ Using Aggregation Function in the Total Sales & Profit.
- ✓ Using Slicers to Filter the Required Data.

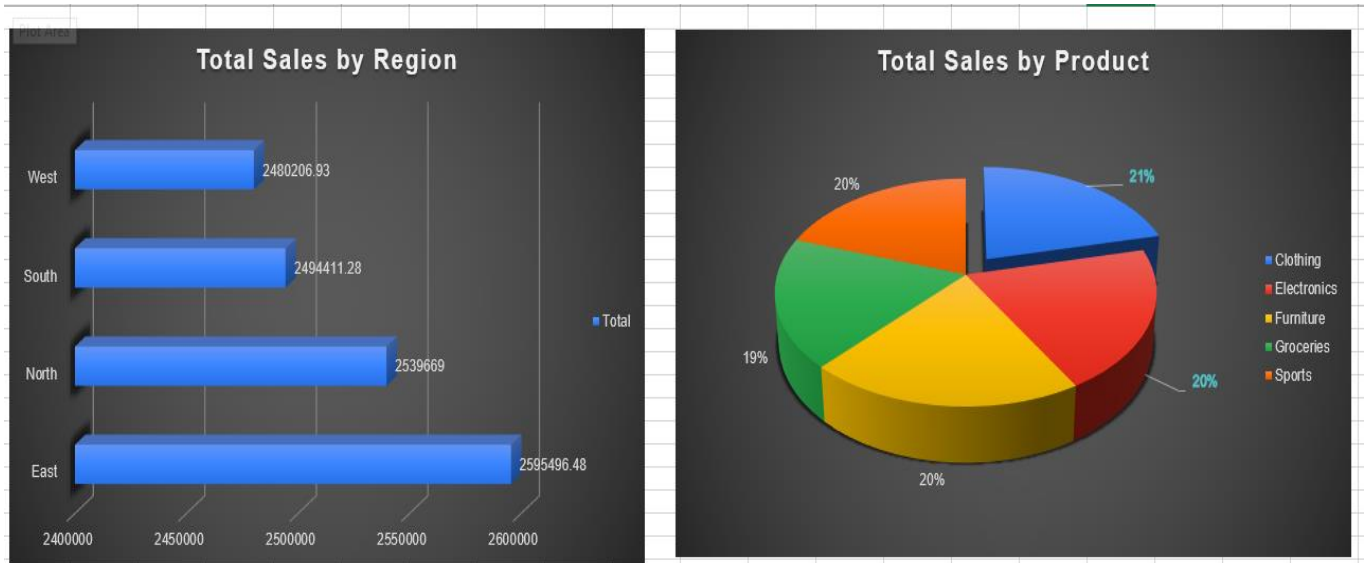
CHARTS			
Region	Sum of Sales Amount	Region	Sum of Profit
East	2595496.48	East	1051660.75
North	2539669	North	970306.07
South	2494411.28	South	1010902.55
West	2480206.93	West	998827.14
Grand Total	10109783.69	Grand Total	4031696.51

Product_C	Sum of Sales Amount	Product_Category	Sum of Profit
Clothing	2156007.58	Clothing	856347.49
Electronics	2047834.93	Electronics	803657.96
Furniture	2049496.16	Furniture	835080.32
Groceries	1864225.06	Groceries	739605.38
Sports	1992219.96	Sports	797005.36
Grand Total	10109783.69	Grand Total	4031696.51

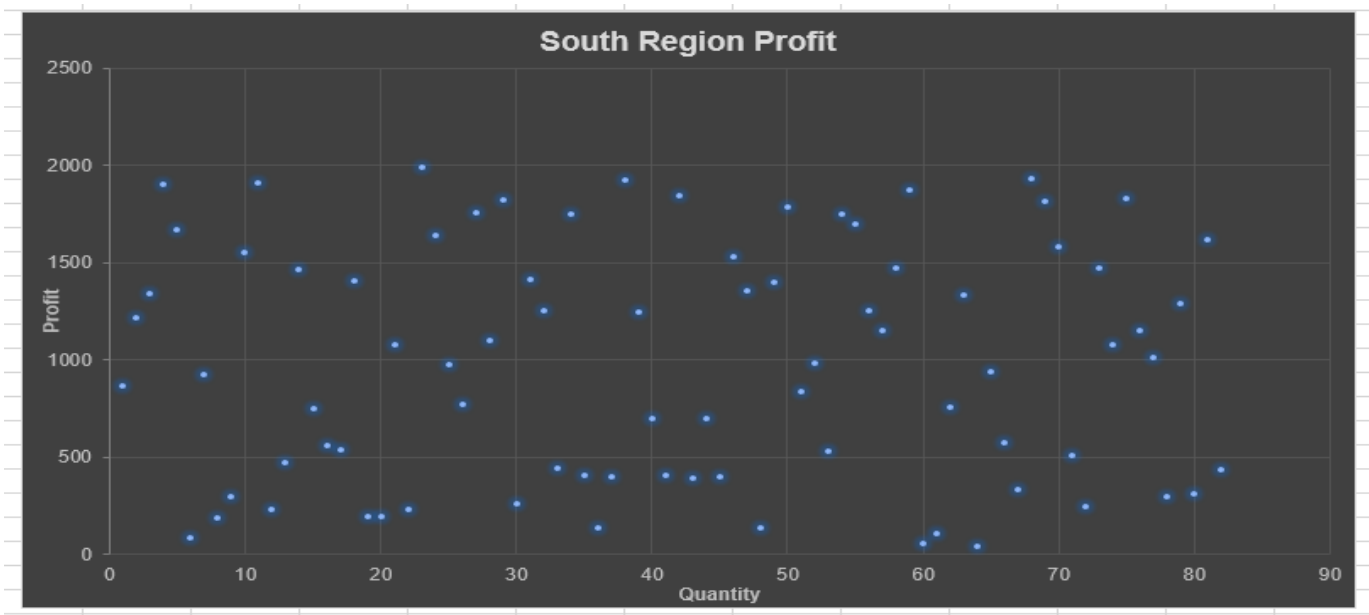
5. Data Visualization

- ✓ Bar Chart for regional sales comparison
- ✓ Pie Chart for product category contribution
- ✓ Stacked Bar Chart for category-wise regional sales
- ✓ Scatter Plot for regression analysis

5.1 Bar Chart & Pie Chart



5.2 Scatter Plot:



5.3 Stacked Chart



6. Dashboard Creation

6.1 Pivot Charts

Pivot Charts were used to visually represent sales and profit data across different regions and product categories. These charts help users easily compare business performance and identify sales trends.

6.2 Slicers and Filters

Slicers and filters were added to make the dashboard interactive. Users can dynamically filter the data based on region, product category, or sales details for better analysis and quick data access.

6.3 Sales Summary

The dashboard provides a summary of total sales generated from different regions. This helps management identify high-performing sales areas and overall business growth.

6.4 Profit Summary

Profit summaries were included to analyze overall profit performance and compare profit margins across product categories and regions.

6.5 Interactive Dashboard

All charts, summaries, and filters were combined into a single interactive dashboard in Microsoft Excel to provide a user-friendly business reporting system.



7. Conditional Formatting

Using conditional formatting to highlight orders with a profit margin greater than 50% or sales amounts above ₹4000.

	A	B	C	D	E	F	G	H	I	J	K
1	Order ID	Order Date	Region	Product Category	Sales Amount	Quantity Sold	Discount (%)	Profit	Average by Region	Average Discount	Average Profit by Region
2	ab899d2f	2023-04-06	North	Groceries	859.29	44	37	1653.87	2573.119554	24.47	983.09
3	73484eb1	2024-04-30	West	Groceries	1546.81	23	33	1932.71	2535.998906	24.47	1,021.30
4	19395924	2024-03-12	East	Groceries	4703.58	31	15	527.47	2507.726068	24.47	1,016.10
5	b3fde3d	2024-02-09	East	Groceries	3377.17	47	3	482.65	2507.726068	24.47	1,016.10
6	6e4bdd51	2024-07-29	South	Clothing	3517.49	6	28	748.72	2494.41128	24.89	1,010.90
7	21d88d69	2023-08-26	East	Furniture	362.55	19	50	314.71	2507.726068	25.23	1,016.10
8	9da2704d	2024-10-07	East	Sports	4811.17	22	18	1372.89	2507.726068	24.81	1,016.10
9	a1bd6815	2023-03-15	West	Electronics	3565.2	22	16	1437.3	2535.998906	24.45	1,021.30
10	0bbd51fc	2023-06-29	East	Clothing	3805.25	19	3	1537.28	2507.726068	24.89	1,016.10
11	8bf7e4b3	2023-03-13	South	Clothing	2178.72	15	24	1696.65	2494.41128	24.89	1,010.90
12	e6cf1255	2024-05-27	West	Groceries	2844.93	34	12	1166.39	2535.998906	24.47	1,021.30

Results:

- ✓ A cleaned and structured dataset ready for analysis.
- ✓ Key insights presented in visualizations, such as the best-performing region and product category.
- ✓ An interactive dashboard summarizing sales performance metrics.
- ✓ A basic regression analysis showing the relationship between discounts and sales

Technical Tags:

Excel, Data Cleaning, Pivot Tables, Charts, Dashboard, Regression Analysis

Skills take away From This Project:

- ✓ Data cleaning and preprocessing in Excel.
- ✓ Searching and filtering data.
- ✓ Working with Excel formulas for calculations and analysis.
- ✓ Creating and interpreting charts (bar chart, pie chart, stacked bar chart).
- ✓ Using Pivot Tables for dynamic data summarization.
- ✓ Building interactive dashboards.
- ✓ Performing basic regression analysis.

Conclusion:

The “Analyzing and Visualizing Regional Sales Performance” project successfully used Microsoft Excel to analyze and visualize sales data effectively. The project helped identify top-performing regions, product categories, and profit trends through charts, Pivot Tables, and dashboards. Data cleaning and regression analysis improved the accuracy and quality of insights. Overall, the project enhanced business decision-making and demonstrated the importance of Excel in retail sales analytics.